

DEN v KC

Drive by Drive Analysis

Kansas City Chiefs 31 - Denver Broncos 10

September 14, 2026 | Arrowhead Stadium | Model learning ledger M01-M88

Purpose: preserve the live possession-by-possession analysis, separate outcome from mechanism, and convert the game into implementable NFL/QB model features.

Evidence labels used in this report: CONFIRMED = structured play-by-play/final stats; LIVE OBSERVATION = broadcast or viewer observation captured during the game; HYPOTHESIS = model-development inference requiring backtest or film/charting validation.

1. Executive Summary

Final result. Kansas City won 31-10. KC finished with 392 total yards at 5.9 yards per play versus Denver's 176 yards at 3.7 per play. Kansas City ran 38 times; Denver ran 15 times. Turnovers were DEN 2, KC 1.

Defining mechanism. Kenneth Walker was the game's primary offensive engine: 23 carries for 173 yards (7.52 YPC) and a 60-yard touchdown. Mahomes finished 15-of-27 for 184 yards, two passing TDs and one INT. The final score therefore overstates how much of Kansas City's advantage came from conventional dropback passing.

Denver's failure mode. Denver repeatedly lost expected value through self-inflicted and system-level events: the opening interception, a catchable Waddle conversion dropped in the third quarter (live observation), delay-of-game penalties, holding, sacks, a strip-sack turnover, an aborted exchange, and red-zone disruption after Mahomes' interception.

Largest modeling lesson. Football projections should be built as conditional state transitions rather than independent box-score averages: early-down outcome -> down/distance -> play-call menu -> defensive response -> protection/coverage -> QB decision -> result -> new state -> coordinator adaptation.

Final Statistical Snapshot

Metric	Denver	Kansas City
Score	10	31
Total yards	176	392
Yards/play	3.7	5.9
Plays	47	67
Rush attempts	15	38
Turnovers	2	1
Possession	26:18	33:42
Penalties-yards	7-52	7-60

Highest-value conclusions for the model

- Early-down success is a causal input to QB difficulty because it changes third-down distance and therefore pressure exposure.
- Raw pass/run ratio is insufficient; distinguish designed run, designed pass, scramble, sack, sneak and kneel, all conditioned on game state.
- Play-caller identity and situational decision quality should be modeled explicitly, including adaptation rate and counterfactual alternatives.
- Player outcomes need causal attribution: QB creation, receiver separation, expected YAC, receiver-created YAC, missed tackles, pressure and protection all require separate credit.
- Turnovers, sacks and penalties require state-dependent value; their cost is a function of field position, down/distance, return, and subsequent capitalization.
- Closeout, terminal and victory-formation snaps must be separated from competitive-state data or they contaminate future tendency and prop projections.
- Official prop-settlement statistics may intentionally differ from the football-performance statistics used for prediction (notably QB kneels).

2. Drive-by-Drive Game Analysis

Chronology reconstructed from the live analysis ledger and structured play-by-play captured during the game. Where a late-game possession advanced faster than the live feed preserved, the report explicitly marks the detail as incomplete rather than inventing a sequence.

1. DEN 1 - Opening possession

Result: Punt/turnover: interception

What happened: Nix's first pass was intercepted deep middle by rookie CB Mansoor Delane. Denver went three plays for negative yardage.

Model learning: Aggressive opening shot immediately exposed interception downside against KC coverage/disguise. Established M07, M26 and negative-event response as live themes.

2. KC 1

Result: Touchdown - Mahomes 15-yard scramble

What happened: Kansas City leaned heavily on Walker and reached the end zone on a Mahomes scramble. Official rushing volume overstated designed-run rate because the scramble originated from a dropback.

Model learning: Validated early-down run success as a pressure suppressant and created the designed-play vs box-score classification problem (M01, M04-M06, M17).

3. DEN 2

Result: Touchdown - Engram 19-yard reception

What happened: Nix rebounded with a 5-for-5 drive. Pat Bryant converted multiple chain-moving targets; Engram scored. Denver shifted toward shorter/intermediate concepts after the opening INT.

Model learning: Strong evidence for play-caller adaptation, high-leverage target share and QB response after negative events (M08-M14).

4. KC 2

Result: Punt - three-and-out

What happened: KC opened with a failed deep attempt, then a short run, then faced 3rd-and-7 and failed to convert. Walker did not touch the ball on the drive.

Model learning: Clean example of failed first-down aggression creating Denver's preferred pressure/down-distance environment (M04, M07, M19).

5. DEN 3

Result: Punt

What happened: Backed up near its own goal line, Denver used Engram/Harvey underneath. A Pat Bryant holding penalty erased a productive play and created a long-yardage state.

Model learning: Expanded disruption modeling beyond QB performance: penalties alter the state tree and passing risk profile (M27-M29).

6. KC 3

Result: Punt

What happened: Walker immediately produced favorable gains and a receiving contribution, but the passing game did not convert the drive. An aborted/fumbled exchange was recovered by KC.

Model learning: Separated early-down success from actual drive continuation and raised offensive continuity/operation as a factor (M30-M33).

7. DEN 4

Result: Punt

What happened: Trautman gained 6, Bryant target incomplete, Waddle caught a short pass on 3rd-and-4 but was stopped before the sticks.

Model learning: A completion can be a defensive win when catch depth and YAC are insufficient. Led to functional-completion and tackling/YAC concepts (M34-M35).

8. KC 4

Result: Punt

What happened: KC remained unable to create passing first downs for much of the first half; first downs came primarily through rushing and Mahomes creation.

Model learning: Reinforced that QB passing stats and team offense are distinct; track first-down source and scramble creation separately (M36-M39).

9. DEN 5

Result: Punt

What happened: Short possession featuring a Troy Franklin rush, a Harvey catch and a Waddle incomplete.

Model learning: Continued evidence of a low-risk short-game structure rather than explosive passing.

10. KC 5

Result: Touchdown - Rice 13-yard reception

What happened: Long late-half TD drive. Walker remained the engine and broke a 28-yard run. Mahomes recovered from a very poor passing start with five straight completions. LIVE OBSERVATION: Rice's TD was materially aided by an easy missed tackle.

Model learning: Created explicit passing-vs-YAC attribution, missed-tackle modeling, fatigue/workload hypotheses and success-weighted play selection (M48-M55).

11. DEN 6

Result: Punt

What happened: With 1:39 left, Nix was sacked, Bryant caught 10, then Sutton target incomplete on third down.

Model learning: Another example of a negative play changing the conversion environment; Bryant continued to emerge as a high-leverage repair target.

12. KC 6

Result: End of half

What happened: Kelce gained 6 and Emmett Johnson ran for 1 before halftime.

Model learning: Halftime state established: KC 14-7; Walker 14 carries/89 yards; Mahomes 8-of-18/62 yards; Nix 12-of-17/98 yards.

13. KC 7 - First possession, 2H

Result: Touchdown - Walker 60-yard run

What happened: On 4th-and-1 KC went quickly with extra linemen and ran Walker outside for 60 yards. LIVE OBSERVATION emphasized quick tempo and heavy personnel.

Model learning: Major evidence for substitution-denial tempo, personnel deception, short-yardage edge vulnerability, fourth-down concept value and halftime adjustment (M56-M60).

14. DEN 7

Result: Punt

What happened: Dobbins opened the drive with three productive runs totaling 25 yards. Denver then took a delay-of-game penalty. LIVE OBSERVATION: Waddle dropped a catchable pass that would have converted the first down.

Model learning: Created catchable-pass attribution, operational-error modeling and expected-drive-value lost by error source (M61-M63).

15. KC 8

Result: Turnover - Mahomes interception

What happened: First snap backed up near its own goal line: deep-middle pass intercepted by Brandon Jones, returned to approximately the KC 5.

Model learning: Powerful contextual-risk example: a volatile deep attempt in an extreme backed-up state despite Walker dominance. Added field-position turnover cost and turnover decomposition (M64-M67).

16. DEN 8

Result: Field goal

What happened: Denver inherited goal-to-go field position after the INT but came away with only three. Holding and a sack/fumble-recovery sequence destroyed the favorable state.

Model learning: Created sudden-change offense/defense, short-field compression, state-dependent negative-play cost and turnover capitalization (M68-M71).

17. KC 9

Result: Field goal

What happened: Walker continued to carry; Kelce broke a 59-yard reception to the Denver 4. Denver then forced a Walker loss, Mahomes scramble for 1 and sack before KC kicked a short field goal.

Model learning: Showed explosive creation can coexist with red-zone defensive compression and that negative-play cost changes sharply near the goal line.

18. DEN 9

Result: Turnover - strip sack

What happened: Late Q3 Nix was sacked by Felix Anudike-Uzomah, fumbled, and KC recovered at the Denver 34.

Model learning: Added sack outcome decomposition and QB ball-security-under-pressure as distinct from ordinary sack susceptibility (M72-M73).

19. KC 10

Result: Touchdown - Walker 2-yard reception

What happened: KC converted the short field. Emmett Johnson gained 25 on a reception, with an additional defensive unnecessary-roughness penalty; Walker finished the drive with a short receiving TD.

Model learning: Showed turnover capitalization, penalty amplification, secondary-RB role substitution and field-position-dependent aggression (M74-M76).

20. DEN 10

Result: Punt - details partially skipped in live capture

What happened: The live feed advanced through this possession before a full play-by-play reconstruction was preserved. Denver punted and KC regained possession.

Model learning: No new M-item assigned; retained to keep the drive ledger complete without inventing details.

21. KC 11

Result: Punt after long clock-drain drive

What happened: With a 31-10 lead, KC leaned on Walker/Emmett Johnson and safe completions, consumed substantial clock, and eventually punted. A late sack mattered far less than it would in a competitive state.

Model learning: Established win-probability-dependent offensive objectives, clock-adjusted play value, closeout personnel redistribution and contamination controls (M77-M80).

22. DEN 11

Result: Turnover on downs

What happened: Poor starting field position after a punt-return holding penalty. Denver had an aborted exchange/fumble it recovered, short completions, and ultimately an incomplete 4th-and-8 throw to Troy Franklin. An initial ruling during the sequence was reversed before the final possession change.

Model learning: Added drive fragility/resilience, special-teams contribution to starting state, and review/overturn state correction (M81-M83).

23. KC 12

Result: Game closeout

What happened: From the Denver 29, KC handed the ball to Emmett Johnson four straight times, then Mahomes took three kneels.

Model learning: Confirmed starter-exposure management and clock-burn efficiency; also showed why late-game RB2 carries and kneels must be separated from normal usage (M84-M87).

24. DEN 12

Result: Terminal kneel

What happened: Nix took a final kneel with the game decided.

Model learning: Created terminal/surrender-state classification and confirmed that these snaps should not train ordinary play-calling or efficiency models (M88).

3. Modeling Architecture Emerging from the Game

Personnel -> play caller selects concept -> defense responds -> protection/routes develop -> QB reads/decides -> pass/scramble/sack -> receiver/YAC/tackle outcome -> new down-and-distance -> coordinators adapt -> repeat.

A. State-transition offense

Represent each snap by down, distance, field position, score, clock, personnel, formation, coverage/front, recent drive state, and current matchup efficiencies. Predict both the play-call branch and the resulting state transition.

B. Separate volume from efficiency

QB attempts are driven by pace, possessions, game state and play-caller preference. QB efficiency is driven by protection, pressure opportunity, coverage, receiver separation, decision quality and YAC. Model them separately before recombining.

C. Play-caller layer

Track actual caller identity, situational fingerprint, adaptation rate, halftime changes, use of discovered matchup edges, and counterfactual decision value. Avoid a single static coach rating.

D. Attribution layer

Decompose outcomes among QB, protection, receiver, scheme, defensive coverage, tackling, penalties, turnover return, and luck. The goal is not merely to explain who received the box-score credit.

E. Data-state hygiene

Exclude or separately label closeout, terminal, kneel, forced fourth-down and other special states so that future tendency models are trained on comparable football decisions.

4. Complete Model Nuance Ledger - M01 to M88

Status key: CORE = strong implementation candidate; BACKTEST = plausible and valuable but needs historical validation; IMPLEMENT = relatively straightforward derived feature; WATCH = collect more evidence; EXTERNAL DATA = requires transcript/film/charting or other non-basic play-by-play data.

Ledger count: CORE 70, BACKTEST 15, WATCH 1, IMPLEMENT 1, EXTERNAL DATA 1 | Total 88

M01 - Designed pass/run vs box-score pass/run [CORE]

Classify designed run, designed pass/dropback, scramble and sack separately. Scrambles/sacks originate from dropbacks and must not inflate designed-run tendency.

M02 - Pass/run ratio must be conditional [CORE]

Condition run/pass choice on down, distance, score, clock, field position, personnel, formation and drive state rather than using a single team rate.

M03 - Opening-script tendency [BACKTEST]

First 10-15 scripted plays can differ materially from rest-of-game behavior. Shrink opening-script evidence before extrapolating full-game usage.

M04 - Early-down success creates downstream QB environment [CORE]

First-down outcome changes second/third-down states, defensive pressure packages and QB difficulty. This was one of the strongest game-wide findings.

M05 - Opponent pressure is not equally exposed every snap [CORE]

Estimate pressure-opportunity rate: how often the offense reaches down/distance and concept states where the defense can actually unleash pressure.

M06 - Scheme can suppress a matchup weakness [CORE]

Run/RPO/quick-game concepts can hide pass-protection weaknesses. Pressure exposure should include dropback depth, protection weakness, rush quality and quick-game suppression.

M07 - Cost of first-down aggression depends on opponent [CORE]

A failed deep shot that creates 2nd-and-10 is more costly against an elite pressure defense than against a passive front.

M08 - High-leverage target share [CORE]

Track targets on third/fourth down, red zone, two-minute and pressure situations separately from raw target share.

M09 - QB trust hierarchy is not raw target share [BACKTEST]

Build Critical Target Rate and Critical Conversion Rate to identify receivers trusted in leverage states; Pat Bryant was a live example.

M10 - Coverage allocation creates secondary-receiver opportunity [CORE]

Receiver opportunity depends on routes, coverage attention, defender matchup and how the defense allocates help, not just receiver talent.

M11 - In-game injuries must be snap-timestamped [CORE]

Historical opponent strength should reflect who was actually on the field for each snap. Delane's injury demonstrated the issue.

M12 - Replacement-player cascade [CORE]

One DB injury can alter safety help, blitz rate and every receiver matchup, not merely the replacement player's assignment.

M13 - QB response after negative events [BACKTEST]

Test performance after interceptions, sacks, fumbles and failed fourth downs. Nix's 5-for-5 TD response after the opening INT is a candidate signal.

M14 - Play-caller adaptation speed [CORE]

Measure how quickly the caller changes concepts after failure, injury, pressure or a newly discovered favorable matchup.

M15 - Drive-level RB workload persistence [BACKTEST]

Do not extrapolate a single high-carry drive linearly; account for rotation, rest and scripted usage on subsequent drives.

M16 - RB rotation purpose [BACKTEST]

Classify substitutions as rest, personnel package, receiving-down role, pass protection, script or late-game preservation.

M17 - QB mobility health signal [BACKTEST]

Live scramble speed/frequency can update a prior on health and available mobility after injury; Mahomes' first-drive TD scramble was a positive signal.

M18 - Health should not be binary [CORE]

Active/inactive is insufficient. Health can affect scramble rate, designed rush, sack avoidance, throw-on-run efficiency and play-caller willingness.

M19 - Third-down distance independent modeling [CORE]

Project the distribution of third-down distance because 3rd-and-2 and 3rd-and-9 expose fundamentally different pressure/coverage environments.

M20 - Receiver usage after defensive personnel change [WATCH]

Track whether an offense intentionally targets the replacement defender or reallocates routes after a secondary injury.

M21 - Explosive-play probability is coverage-state dependent [CORE]

Box count and safety shell change ADOT, route access and explosive probability; successful rushing can force structural passing opportunities.

M22 - Game-script pass-rate elasticity [CORE]

Pass rate should depend on score x quarter x time remaining x win probability x matchup efficiency, not score differential alone.

M23 - Successful short passing can function like rushing [IMPLEMENT]

Screens, quick throws and RB outlets can be modeled as run-extension concepts when they serve similar state-control purposes.

M24 - Personnel-specific defensive coverage schema [CORE]

Team-level man/zone rates are insufficient when personnel changes. Coverage expectations should condition on available defenders and package.

M25 - Bayesian shrinkage for live updating [CORE]

Do not overreact to one drive, target cluster or explosive play. Update priors more strongly only when the underlying mechanism repeats.

M26 - Outcome vs mechanism classification [CORE]

Update the model primarily when the causal mechanism changes, not simply because a TD, INT or explosive outcome occurred.

M27 - Penalty/drive disruption state [CORE]

Track penalties, sacks, TFLs, aborted snaps and negative runs as nonlinear state disruptions that change future play menus and predictability.

M28 - RB receiving role as pressure/long-yardage solution [BACKTEST]

Separate screens, checkdowns, angle routes and option routes; receiving work can replace carries when the offense becomes pass-heavier.

M29 - Starting field position affects passing risk profile [CORE]

Backed-up field position may change ADOT, deep-shot frequency and negative-play avoidance more than raw pass rate.

M30 - Drive continuation efficiency [CORE]

Favorable early-down states only matter if the offense actually converts and stays on the field. Model continuation probability explicitly.

M31 - Separate QB volume from QB efficiency [CORE]

Attempts arise from possessions/pace/pass rate/game state; efficiency arises from protection/coverage/pressure/QB/receiver execution.

M32 - Offensive touch concentration [BACKTEST]

Model carries plus targets and how workload shifts among rushes, checkdowns and screens when game state changes.

M33 - Offensive operation/continuity [BACKTEST]

OL/center and system continuity may affect false starts, aborted exchanges, protection busts and delay penalties, especially early season.

M34 - Completion quality / functional completion [CORE]

Use target depth relative to sticks and YAC requirement. A completion short of the sticks can be a defensive win.

M35 - Defensive rally-and-tackle ability [BACKTEST]

Track YAC allowed, YAC over expected, third-down YAC conversions and tackle success after short completions.

M36 - QB statistical production is not team offensive production [CORE]

Route passing production through field position, conversions, red zone and scoring efficiency rather than mapping QB yards directly to points.

M37 - First-down generation by play type [CORE]

Track first downs created by rush, completed pass, scramble and penalty; KC's early first downs came almost entirely outside conventional passing.

M38 - Broadcast intelligence extraction [EXTERNAL DATA]

Capture timestamped announcer/sideline information on injuries, personnel, scheme, tendencies and coaching comments, then validate against structured data.

M39 - QB creation can rescue failed passing structure [CORE]

A designed dropback can end in completion, scramble or sack. Successful scrambles are QB creation, not evidence that the caller chose a run.

M40 - Offensive mechanism diversity [CORE]

Count independent successful mechanisms (inside/outside run, quick pass, intermediate, scramble, etc.). Offenses with one working mechanism are more fragile.

M41 - Receiver-yardage decomposition [CORE]

Model routes -> target probability -> catch probability -> yards at catch -> YAC rather than forecasting receiving yards as a single block.

M42 - Play-caller identity [CORE]

Track the actual play caller rather than only HC/OC title; shrink tendencies when caller changes.

M43 - Situational play-calling fingerprint [CORE]

Profile caller behavior in short yardage, red zone, backed-up, after turnovers/explosives, two-minute and long-yardage states.

M44 - Play-call quality separate from play result [CORE]

A good call can fail and a bad call can succeed. Evaluate the chosen concept against expected alternatives rather than grading by outcome.

M45 - Play-caller adaptation rate [CORE]

Measure how quickly the caller changes after failed concepts, injuries, pressure and repeated defensive looks.

M46 - Play caller x QB style interaction [CORE]

Model QB talent x caller fit x defensive matchup, not simply additive QB and coach ratings.

M47 - Offensive caller vs defensive caller matchup [CORE]

Estimate how a caller performs against defensive archetypes and specific coordinator tendencies.

M48 - Passing creation vs YAC creation [CORE]

Separate air yards at catch, expected YAC, receiver-created YAC and defensive-error YAC.

M49 - Missed-tackle probability / tackling quality [BACKTEST]

Use defender/team missed-tackle rate, tackle success, yards after contact and YACOE allowed, conditioned on receiver type.

M50 - Explosive-play causal classification [CORE]

Classify explosives as QB-created, separation-created, scheme-created, YAC-created or defensive-error-created.

M51 - Drive continuity state after conversions [BACKTEST]

A conversion may improve short-term efficiency through fatigue, personnel trapping, information gain and sequencing. Treat as continuity, not vague momentum.

M52 - Cumulative defensive workload/fatigue [BACKTEST]

Track recent defensive snaps, rush-defense snaps, drive length and substitution opportunity as predictors of tackling and fit deterioration.

M53 - Success-weighted play selection [CORE]

Model whether callers appropriately increase usage of concepts that are working in the specific matchup, with Bayesian protection against small samples.

M54 - Counterfactual play-call value [CORE]

Estimate chosen concept expected success/value minus plausible alternative concepts from the same state.

M55 - Result luck vs decision quality [CORE]

Do not let a lucky TD or unlucky drop retrospectively validate or invalidate the underlying decision; Rice's missed-tackle TD was the live example.

M56 - Substitution-denial tempo [CORE]

Track quick snaps/no-huddle used specifically to trap unfavorable defensive personnel, not just generic plays per minute.

M57 - Personnel package as information and deception [CORE]

Heavy personnel creates defensive expectations; tendency breakers can exploit the response generated by the package itself.

M58 - Short-yardage front compression vs edge vulnerability [CORE]

Condensed interior fits can make outside runs more attractive; model box density, front width, edge alignment and run direction.

M59 - Fourth-down decision vs fourth-down concept [CORE]

Separate the decision to go from the chosen play. Refine further into discretionary fourth downs versus forced late-game attempts.

M60 - Halftime Adjustment Delta [CORE]

Compare first-half strategy to the first two second-half drives: run rate, target share, box structure, pressure and concept use.

M61 - Catchable-pass attribution / receiver error [CORE]

Classify incompletions as QB miss, drop, breakup, throwaway, pressure-caused miss, miscommunication or batted pass.

M62 - Pre-snap operational efficiency [CORE]

Track delay of game, false start, illegal formation/shift, substitution issues and aborted exchanges, with responsibility attribution where possible.

M63 - Expected drive value lost by error source [CORE]

Decompose drive EV gained/lost among QB, receiver, OL/protection, play caller, penalties/operations and defense.

M64 - Contextual risk budget [CORE]

Optimal aggression depends on score, field position, clock, opponent and available low-variance alternatives. Evaluate full downside, not upside only.

M65 - Field-position-adjusted turnover cost [CORE]

Turnovers should be valued by location, return, resulting field position and opponent expected points rather than a flat -1 count.

M66 - Deep-attempt downside distribution [CORE]

Deep throws combine explosive upside with lower completion, longer protection and interception/return risk; model the full distribution.

M67 - Turnover outcome decomposition [CORE]

For INTs/fumbles separate creation, catch/recovery, return and post-turnover field position; do not assign all downstream return value to the QB.

M68 - Sudden-change offense and defense [CORE]

Treat possessions immediately after turnovers as their own state because field position, substitutions and defensive behavior change abruptly.

M69 - Short-field defensive compression [CORE]

Near the goal line vertical space collapses, route windows shrink and defenders can play more aggressively. Red-zone coverage is a distinct environment.

M70 - State-dependent negative-play cost [CORE]

A sack/penalty's cost should include the change in TD/scoring probability, not just yards lost; red-zone negative plays are especially expensive.

M71 - Turnover capitalization rate [CORE]

Measure expected points from inherited post-turnover state versus actual points. Separate takeaway creation from offensive conversion.

M72 - Sack outcome decomposition [CORE]

Separate pressure, sack, strip probability, recovery and resulting field position; strip-sack ability differs from ordinary sack ability.

M73 - QB ball security under pressure [CORE]

Track fumbles per pressure/sack, strip-sack rate, extension behavior and recovery luck independently from interception propensity.

M74 - Penalty amplification after successful plays [CORE]

Penalty value should be EPA/state based. A foul following an explosive gain can have multiplicative scoring impact.

M75 - Secondary RB receiving / role substitution [BACKTEST]

The offense can preserve backfield receiving stress through a second RB; individual usage changes do not necessarily remove the structural threat.

M76 - Field-position-dependent offensive aggression [CORE]

Aggressive calls have different downside asymmetry in plus territory versus backed-up territory; field position should directly alter risk tolerance.

M77 - Win-probability-dependent offensive objective [CORE]

As win probability rises, objective shifts from maximizing EPA toward first downs, clock, ball security, field position and player preservation.

M78 - Clock-adjusted play value [CORE]

Late-game play value should include clock consumed/stopped conditional on win probability, not EPA alone.

M79 - Late-game personnel/workload redistribution [CORE]

Model RB1/RB2 usage share by quarter, lead size and win probability; large leads may send closeout carries to backups.

M80 - Closeout-state contamination of historical stats [CORE]

Label competitive, closeout, comeback and end-half states. Do not let terminal run-heavy or backup usage distort neutral tendencies.

M81 - Drive fragility / disruption resilience [CORE]

Estimate the probability an offense can survive one negative event. Low-explosive offenses may collapse after a single penalty/sack/aborted snap.

M82 - Special-teams contribution to offensive starting state [BACKTEST]

Starting field position is partly created by punt/kick quality, returns and special-teams penalties; include hidden-yardage effects in team/QB context.

M83 - Review/overtake state correction [CORE]

Live systems must mark rulings provisional until replay/penalty resolution. Do not finalize possession or feature updates before adjudication is stable.

M84 - Starter exposure / injury-risk management [CORE]

As win probability approaches 100%, coaches may trade marginal football value for reduced contact and injury exposure to high-value starters.

M85 - Clock-burn efficiency per snap [CORE]

In closeout states measure seconds removed, first-down probability and turnover risk. Mundane in-bounds gains can be strategically optimal.

M86 - Victory-formation plays excluded from normal tendencies [CORE]

Kneels are clock-management events, not designed QB rushing. Exclude from run rate, YPC and normal play-caller tendency estimates.

M87 - Predictive football stats vs official prop-settlement stats [CORE]

Use football-normalized stats for prediction but project official scoring separately when settlement includes kneel yards or other bookkeeping effects.

M88 - Terminal/surrender-state classification [CORE]

When both teams have abandoned competitive objectives, exclude the snaps from normal efficiency, aggressiveness, target-share and tendency learning.

5. Engineering Roadmap Derived from the Ledger

Phase 1 - Implement from standard play-by-play immediately

- M01/M02 designed-play normalization; M04/M19 down-distance state engine; M22 game-script elasticity; M27 disruption events; M31 QB volume vs efficiency; M37 first-down source; M59 fourth-down state; M64-M71 turnover/state value; M77-M80 closeout classification; M83 ruling finalization; M86-M88 kneel/terminal cleanup.

Phase 2 - Backtest from richer public tracking / charting data

- M05/M06 pressure opportunity and suppression; M08-M10 leverage/coverage target share; M21/M24 shell and personnel state; M34/M35 functional completion and tackling; M41/M48-M50 air/YAC/explosive attribution; M49 missed-tackle probability.

Phase 3 - Play-caller/coordinator layer

- M14 and M42-M47 caller identity, fingerprints, adaptation and coordinator matchup; M53/M54 success-weighted and counterfactual selection; M56-M60 tempo, personnel deception, fourth-down concept and halftime delta.

Phase 4 - Operations, health and live intelligence

- M11-M18 injury/health/rotation; M33 operation continuity; M38 transcript/broadcast intelligence; M61-M63 attribution of drops and operational errors; M72-M76 pressure ball security and penalty interaction.

Phase 5 - Player-prop settlement layer

- M79 RB1/RB2 late-game usage; M84 starter exposure; M85 clock burn; M86 kneel removal for football performance; M87 re-add expected official-stat kneel effects where sportsbook settlement requires them.

Postgame film / transcript verification list

- Review Nix opening interception: coverage shell, leverage, route concept and decision quality.
- Confirm Mansoor Delane injury timing, return status and Denver target behavior after the personnel change.
- Review KC 4th-and-1 Walker 60-yard TD: substitution timing, defensive front width, extra OL alignment and edge fit.
- Grade the Waddle third-quarter incompleteness: true drop/catch probability versus throw quality.
- Review Mahomes backed-up interception: Denver box/shell and whether the deep call had a valid counterfactual justification.
- Grade Rice TD YAC: expected tackle probability, defender error and receiver-created yards.
- Review Denver goal-to-go series after Mahomes INT: play designs, protection and expected points lost.
- Attribute Nix aborted exchange and strip-sack between QB/center/protection/QB extension where film allows.
- Extract announcer/sideline notes on play calling, injuries, personnel and adjustments and timestamp them to snaps.

6. Closing Model Principle

The final score is an outcome. The model should learn the sequence of states and decisions that made the outcome probable.

The central engineering implication from DEN-KC is that static player averages are too lossy for serious prediction. The next-generation QB/NFL model should explicitly represent state transitions, play-caller decisions, pressure opportunity, personnel, attribution, disruption, adaptation and terminal-state data hygiene. M01-M88 is the working specification produced by this game.

Source note: compiled from live structured NFL game data captured during the DEN @ KC game, final game statistics, and user-provided live broadcast observations during the analysis session. Broadcast observations are labeled as such and should be verified against replay/gamebook before being used as ground-truth training labels.